

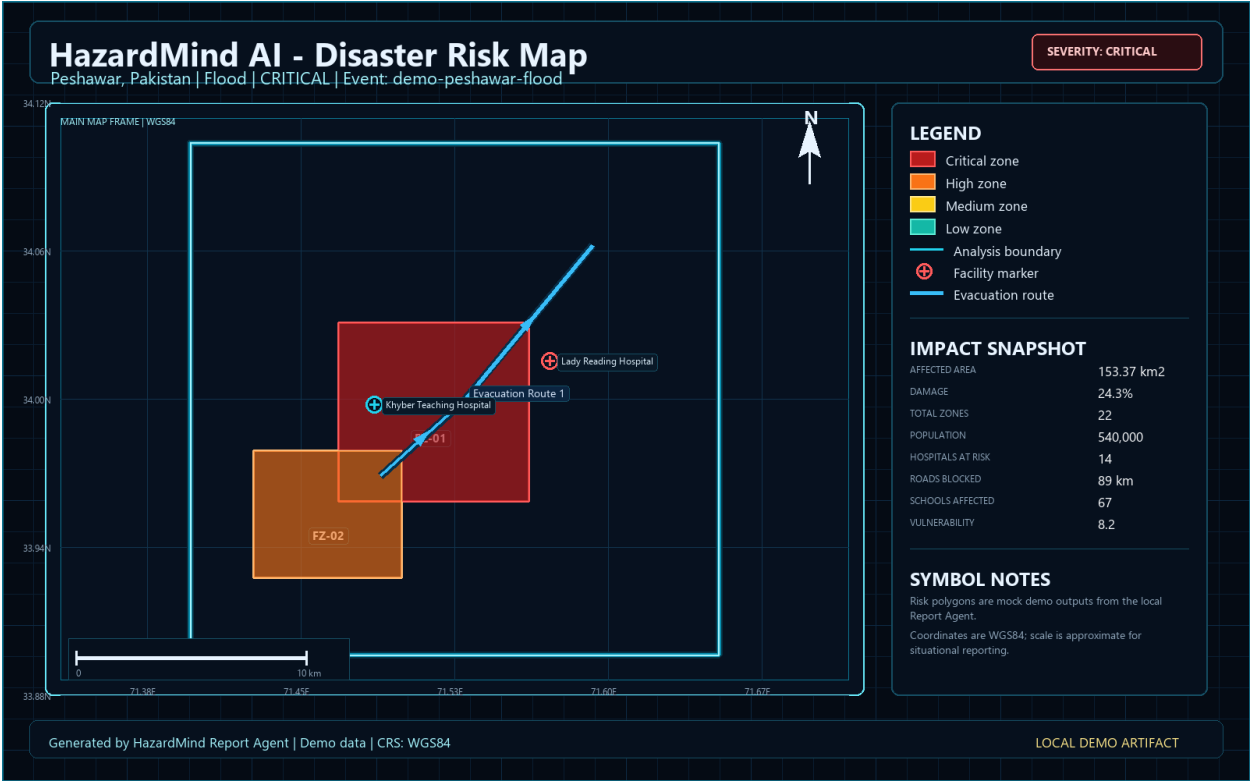
HazardMind AI Executive Disaster Report

Event ID	demo-peshawar-flood
Location	Peshawar, Pakistan
Hazard Type	Flood
Overall Severity	CRITICAL
Satellite Type	sentinel-1
Satellite Reason	cloud_cover_above_30_percent_sar_selected

Operational Statistics

Affected Area	153.37 km2
Damage Percent	24.3%
Total Zones	22
Population Affected	540,000
Hospitals at Risk	14
Roads Blocked	89 km
Schools Affected	67
Vulnerability Score	8.2

Generated Risk Map



Executive Summary

Peshawar, Pakistan requires immediate escalation for a CRITICAL flood event affecting 540,000 people. Critical decisions are needed on evacuation routing, hospital continuity, road clearance, and shelter activation.

Intelligence Assessment

Criticality	critical
Overall Confidence	91%
Escalation Required	True
Rationale	Critical flood in Peshawar with 153 km² inundated, 540,000 people at risk, 14 hospitals threatened, and only 67 schools affected.
Key Decisions	Authorize immediate evacuation support for critical flood zones.; Prioritize hospital access and medical continuity.
Human Review Required	False

Map Narrative

The static risk map identifies a critical flood event in Peshawar, covering 153.37 km² (24.3% damage extent). High-confidence SAR imagery confirms significant inundation affecting 540,000 people, with critical infrastructure threats and severe logistical bottlenecks due to limited evacuation routes.

- 153.37 km² of total inundated area identified via SAR VV/VH ratio
- 89 km of road networks blocked, severely limiting mobility
- High vulnerability score of 8.2 indicating poor local resilience

- Single remaining evacuation route creating a critical operational bottleneck

Priority Timeline

Next 6 Hours	Evacuate critical flood zones FZ-01 and FZ-02 immediately; Secure access routes to Lady Reading Hospital
Next 24 Hours	Clear 89km of blocked roads for logistics flow; Establish temporary shelters for 540,000 affected persons; Distribute emergency supplies
Next 72 Hours	Conduct field validation of satellite damage assessments; Restore critical utility services to affected schools and clinics
Resources	Heavy machinery for road clearance; Emergency medical supplies and mobile clinics; High-capacity water purification units
Coordination	Manage bottleneck at single available evacuation route; Synchronize satellite data with field reconnaissance reports

Anomalies and Warnings

- MEDIUM: Missing pdf_url and map_url in report object Handling: Verify output generation pipeline and re-trigger PDF/Map exports
- LOW: Single evacuation route for 540,000 affected people creates extreme operational bottleneck Handling: Validate route count via ground truth or alternative GIS datasets
- GeoJSON artifact URL is not available in local demo mode.
- Map URL is generated later by the local artifact step.
- Single evacuation route for 540,000 people creates severe logistical bottleneck; verify route count via ground truth.
- SAR-derived flood extent and damage metrics require immediate field validation for operational accuracy.
- Optical validation unavailable due to 42% cloud cover; interpret SAR results with inherent ambiguity.

Quality Check

Status	not_ready
Event Id Present	True
Satellite Data Present	True
Hazard Data Present	True
Impact Data Present	True
Map Artifacts Present	True
Recommendations Present	True
Confidence Above Threshold	True
Warnings	GeoJSON artifact URL is not available in local demo mode.; Map URL is generated later by the local artifact step.
Blocking Issues	Critical report assets (pdf_url, map_url) are missing; generate and attach before distribution.

Band-Ready Final Message

HazardMind Report Agent complete for demo-peshawar-flood (Peshawar, Pakistan, Flood). Criticality: critical; confidence: 91%. Summary: Peshawar, Pakistan requires immediate escalation for a CRITICAL flood event affecting 540,000 people. Critical decisions are needed on evacuation routing, hospital continuity, road clearance, and shelter activation.

Detailed Incident Analysis

SAR analysis using Sentinel-1 VV/VH ratio confirms widespread flooding across Peshawar, with a mean index of 0.24 and 24.3% damage extent covering 153.37 km². Zones FZ-01 and FZ-02 exhibit critical deep water and high water severity. Flood risk is assessed as CRITICAL with a confidence score of 0.91. Approximately 540,000 people are affected; 14 hospitals including Lady Reading Hospital face high risk, 67 schools are impacted, and 89 km of roads are blocked. Vulnerability score is 8.2 out of 10. The single available evacuation route severely constrains response capacity.

Technical Analysis

Sentinel-1 SAR was selected because cloud cover is 42%. The analysis uses SAR VV/VH ratio with mean value 0.24 and identifies 22 zones. Damage is estimated at 24.3%, with 89 km of roads blocked and a vulnerability score of 8.2.

Recommendations

- Prioritize evacuation in critical flood zones.
- Deploy emergency medical support near hospitals at risk.
- Clear blocked road corridors for rescue access.
- Open temporary shelters for displaced families.
- Monitor flood expansion using updated satellite imagery.
- Evacuate critical flood zones FZ-01 and FZ-02 immediately
- Secure access routes to Lady Reading Hospital
- Deploy emergency search and rescue teams to isolated zones
- Heavy machinery for road clearance
- Emergency medical supplies and mobile clinics

Response Priorities

- Activate evacuation support for FZ-01 and FZ-02.
- Protect hospital access and stage medical surge teams near high-risk facilities.
- Clear blocked road corridors needed for rescue and logistics movement.

Assumptions

- Satellite, hazard, and impact values are local demo data until upstream agents publish live outputs.
- Risk zone geometries are simplified mock polygons for dashboard and report demonstration.
- Facility and route coordinates are approximate and intended for prototype visualization.

Limitations

- No live Band messages are consumed in this local demo run.
- No field validation, hydrologic model, or live basemap tiles are used for artifact generation.
- Impact estimates should be treated as decision-support indicators, not official counts.

Model Source Note

Detailed report generated using Featherless Kimi K2.6. Executive summary generated using AI/ML API. Intelligence sources: criticality: featherless:deepseek-v4-pro, anomaly_check: featherless:gemma-4-31b-it, map_narrative: featherless:gemma-4-31b-it, priority_recommendations: featherless:gemma-4-31b-it, decision_brief: deterministic_fallback, quality_check: hybrid:deterministic+featherless:qwen-3.6-35b-a3b, band_ready_message: template+intelligence

Detailed Report	featherless:composite
Executive Summary	deterministic_fallback
Fallback Used	False
Featherless Model	moonshotai/Kimi-K2.6
Criticality	featherless:deepseek-v4-pro
Map Narrative	featherless:gemma-4-31b-it
Priority Timeline	featherless:gemma-4-31b-it
Quality Check	hybrid:deterministic+featherless:qwen-3.6-35b-a3b

Agent Log / Trace

Agent	Status	Timestamp	Message
hazardmind-report	complete	2026-06-13T18:03:00Z	Report Agent received event context
hazardmind-report	complete	2026-06-13T18:03:20Z	Featherless composite generated detailed disaster report
hazardmind-report	complete	2026-06-13T18:03:40Z	AI/ML generated executive summary
hazardmind-report	complete	2026-06-13T18:03:45Z	Criticality assessed
hazardmind-report	complete	2026-06-13T18:03:50Z	Anomalies checked
hazardmind-report	complete	2026-06-13T18:03:55Z	Map narrative generated
hazardmind-report	complete	2026-06-13T18:04:00Z	Priority timeline generated
hazardmind-report	complete	2026-06-13T18:04:05Z	Decision brief generated with Opus
hazardmind-report	complete	2026-06-13T18:04:10Z	Quality check completed
hazardmind-report	complete	2026-06-13T18:04:15Z	Band-ready final message prepared
hazardmind-report	complete	2026-06-13T18:04:00Z	Static cartography map generated locally